Session 1 | Probability I

Exercise sheet

Work through these in order: they start gently and become more challenging toward the end, including a few famous probability paradoxes. Show your reasoning and your calculations, most answers are short. If a question defeats you, write "I don't know" *and we will work it out together.*

Exercise 1 *Imagine two independent noisy channels, each reporting an integer from* 1 *to* 6*, like two dice, or two coarse spike-count sensors.*

Two fair dice are rolled. Compute:

(a) $\mathbb{P}$(the sum equals 7);

(b) $\mathbb{P}$(the sum equals 7 or 11);

(c) $\mathbb{P}$(at least one die shows a 6).

Exercise 2 *In a rapid serial visual presentation, a faint target may appear on each briefly-flashed frame.*

A target appears on each of 5 independent frames with probability 0.15. What is the probability that *at least one* target appears in the sequence? *(Use the complement.)*

Exercise 3 Among 200 participants in a study, 120 are bilingual, 90 are trained musicians, and 50 are both. A participant is chosen at random. Find:

(a) $\mathbb{P}$(bilingual or musician);

(b) $\mathbb{P}$(neither);

(c) $\mathbb{P}$(exactly one of the two).

Exercise 4 *Stimulus sets are often assembled by random draws from a larger pool.*

An experimenter randomly selects 3 images from a pool of 10, of which 4 depict faces. Compute:

(a) the probability that all 3 selected images are faces;

(b) the probability that exactly one of the 3 is a face.

Exercise 5 **The birthday paradox.** Assume 365 equally likely birthdays.

(a) For a group of n people, write an expression for the probability that *no two* share a birthday.

(b) Evaluate the probability that *at least two* share a birthday in a small group of $n = 4$.

(c) For a class of $n = 23$ the probability of a shared birthday is about 0.507. Explain in one or two sentences why this is so much higher than most people expect. *(Hint: how many pairs of people are there?)*

Exercise 6 *A signal-detection setup: "old" vs. "new" stimuli and the participant's "old"/"new" response.*

Out of 200 memory trials: of the 100 genuinely *old* items, the subject called 80 "old"; of the 100 *new* items, the subject called 30 "old". Compute:

(a) the hit rate $\mathbb{P}(\text{says old} \mid \text{old})$;

(b) the false-alarm rate $\mathbb{P}(\text{says old} \mid \text{new})$;

(c) $\mathbb{P}(\text{item was old} \mid \text{says old})$.

Exercise 7 Two neurons respond to a stimulus. Neuron A fires on 40% of trials and neuron B on 30% of trials, and (for now) they fire independently.

(a) Compute $\mathbb{P}(\text{both fire})$ and $\mathbb{P}(\text{at least one fires})$.

(b) Suppose instead you measured $\mathbb{P}(\text{both fire}) = 0.20$. Are the two neurons still independent? What does the discrepancy suggest?

Exercise 8 *A perceptual task mixes trials of different difficulty.*

Trials are 50% easy (accuracy 0.95), 30% medium (accuracy 0.80), and 20% hard (accuracy 0.55). If a trial is drawn at random, what is the overall probability of a correct response? *(Use the law of total probability.)*

Exercise 9 **The false-positive "paradox".** A test for a neurological condition that is present in 2% of the population has a sensitivity of 95% (it correctly flags true cases) and a specificity of 90% (so it wrongly flags 10% of healthy people). A randomly chosen person tests positive.

(a) Compute the probability that they actually have the condition.

(b) In one sentence, explain why the answer is far below the test's 95% sensitivity.

Exercise 10 **The Monty Hall problem.** In a decision task with three doors, a reward sits behind one. You choose door 1. The host, who knows where the reward is, opens door 3, revealing nothing, and offers you the chance to switch to door 2.

(a) What is your probability of winning if you *stay* with door 1?

(b) What is it if you *switch* to door 2?

(c) Explain briefly why switching helps, in terms of the information the host's choice provides.

Exercise 11 *A two-alternative forced-choice (2AFC) task, modelled as independent Bernoulli trials.*

A subject performs at 75% accuracy over $n = 12$ independent trials; let X be the number of correct responses.

(a) Name the distribution of X and give its parameters.

(b) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

(c) Compute the probability of a perfect score, $\mathbb{P}(X = 12)$.

Exercise 12 *Cortical neurons are often modelled as Poisson spike generators.*

A neuron fires as a Poisson process with mean $\lambda = 4$ spikes per 100 ms window.

(a) Compute the probability of exactly 2 spikes in a window.

(b) Compute the probability of *no* spikes in a window.

(c) State the variance of the spike count, and explain what "variance equals mean" would let you check in real data.

Exercise 13 Reaction times are approximately Normal with mean 600 ms and standard deviation 80 ms.

(a) Compute the z-score of a 760 ms response.

(b) Using the 68–95–99.7 rule, roughly what fraction of responses are *slower* than 760 ms?

(c) Roughly what fraction fall between 520 and 680 ms?

Exercise 14 *Evidence accumulation: a detector provides one piece of evidence about whether a stimulus is a target.*

Before any evidence, the odds that a stimulus is a target are $1 : 4$. A neural detector fires, and firing is 6 times more likely for targets than for non-targets (a likelihood ratio of 6).

(a) Compute the posterior odds that the stimulus is a target.

(b) Convert these odds to a posterior probability.

Exercise 15 **The two-child problem.** A family has two children (each equally likely a boy or a girl, independently).

(a) Given that *at least one* child is a girl, what is the probability that *both* are girls?

(b) Given that *the elder* child is a girl, what is the probability that both are girls?

(c) Explain why the two answers differ, even though both involve "a girl".